

Membuat Code Salah Satu Use Case

Use Case Scenario

Basic Flow (Main Success Scenario):

- User membuka halaman login
- Sistem menampilkan form login (email dan password)
- User memasukkan email dan password
- User menekan tombol "Login"
- Sistem memvalidasi input
- Sistem memeriksa kredensial ke database
- Sistem memverifikasi password
- Sistem membuat session user
- Sistem mengarahkan user ke dashboard
- Use case selesai

Alternative Flow:

A1. Email tidak ditemukan:

Pada langkah 6, jika email tidak terdaftar:

Sistem menampilkan pesan "Email tidak terdaftar"

Kembali ke langkah 2

A2. Password salah:

Pada langkah 7, jika password tidak cocok:

Sistem menampilkan pesan "Password salah"

Kembali ke langkah 2

A3. Input kosong:

Pada langkah 5, jika ada field kosong:

Sistem menampilkan pesan "Email dan password harus diisi"

Kembali ke langkah 2

Implementasinya

Model – UserModel.php

```
<?php
```

```
class UserModel {
```

```
    private $conn;
```

```
    private $table = 'users';
```

```
    public function __construct($db) {
```

```
        $this->conn = $db;
```

```
    }
```

```
    // Login user
```

```
    public function login($email, $password) {
```

```
        $query = "SELECT id_user, nama, email, password, role
```

```
                FROM " . $this->table . "
```

```
                WHERE email = :email";
```

```
        $stmt = $this->conn->prepare($query);
```

```
        $stmt->bindParam(':email', $email);
```

```
        $stmt->execute();
```

```
        if ($stmt->rowCount() > 0) {
```

```
            $row = $stmt->fetch(PDO::FETCH_ASSOC);
```

```

        // Verify password
        if (password_verify($password, $row['password'])) {
            return new User(
                $row['id_user'],
                $row['nama'],
                $row['email'],
                $row['password'],
                $row['role']
            );
        }
    }

    return false;
}

// Check if email exists
public function emailExists($email) {
    $query = "SELECT id_user FROM " . $this->table . " WHERE email = :email";
    $stmt = $this->conn->prepare($query);
    $stmt->bindParam(':email', $email);
    $stmt->execute();
    return $stmt->rowCount() > 0;
}
}
?>

```

Controller - LoginController.php

<?php

```
require_once 'User.php';
```

```
require_once 'UserModel.php';
```

```
class LoginController {
```

```
    private $userModel;
```

```
    public function __construct($db) {
```

```
        $this->userModel = new UserModel($db);
```

```
    }
```

```
    public function processLogin($email, $password) {
```

```
        $errors = [];
```

```
        // Validation
```

```
        if (empty($email) || empty($password)) {
```

```
            $errors[] = "Email dan password harus diisi";
```

```
            return ['success' => false, 'errors' => $errors];
```

```
        }
```

```
        if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
```

```
            $errors[] = "Format email tidak valid";
```

```
            return ['success' => false, 'errors' => $errors];
```

```
        }
```

```
        // Attempt login
```

```
        $user = $this->userModel->login($email, $password);
```

```

if ($user) {
    // Set session

    $_SESSION['user_id'] = $user->getIdUser();
    $_SESSION['user_nama'] = $user->getNama();
    $_SESSION['user_email'] = $user->getEmail();
    $_SESSION['user_role'] = $user->getRole();
    $_SESSION['logged_in'] = true;

    return [
        'success' => true,
        'user' => $user,
        'message' => "Login berhasil, selamat datang " . $user->getNama()
    ];
} else {
    // Check specific error
    if (!$this->userModel->emailExists($email)) {
        $errors[] = "Email tidak terdaftar";
    } else {
        $errors[] = "Password salah";
    }

    return ['success' => false, 'errors' => $errors];
}
}

```

```

public function logout() {
    session_destroy();
}

```

```

        return ['success' => true, 'message' => "Logout berhasil"];
    }
}
?>

View - login.php

<?php
session_start();

// Redirect if already logged in
if (isset($_SESSION['logged_in']) && $_SESSION['logged_in'] === true) {
    header("Location: dashboard.php");
    exit();
}

$errors = [];
$email = "";

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    require_once 'config/database.php';
    require_once 'LoginController.php';

    $database = new Database();
    $db = $database->getConnection();
    $loginController = new LoginController($db);

    $email = $_POST['email'] ?? "";

```

```
$password = $_POST['password'] ?? '';
```

```
$result = $loginController->processLogin($email, $password);
```

```
if ($result['success']) {
```

```
    header("Location: dashboard.php");
```

```
    exit();
```

```
} else {
```

```
    $errors = $result['errors'];
```

```
}
```

```
}
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html lang="id">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Login - Sistem Informasi</title>
```

```
    <style>
```

```
        * { margin: 0; padding: 0; box-sizing: border-box; }
```

```
        body {
```

```
            font-family: Arial, sans-serif;
```

```
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
```

```
            min-height: 100vh;
```

```
            display: flex;
```

```
            align-items: center;
```

```
        justify-content: center;
    }

    .login-container {
        background: white;
        padding: 40px;
        border-radius: 10px;
        box-shadow: 0 10px 25px rgba(0,0,0,0.2);
        width: 100%;
        max-width: 400px;
    }

    .login-header {
        text-align: center;
        margin-bottom: 30px;
    }

    .login-header h2 {
        color: #333;
        margin-bottom: 10px;
    }

    .form-group {
        margin-bottom: 20px;
    }

    .form-group label {
        display: block;
        margin-bottom: 5px;
        color: #555;
        font-weight: bold;
    }
```



```
.form-group input {  
    width: 100%;  
    padding: 12px;  
    border: 1px solid #ddd;  
    border-radius: 5px;  
    font-size: 14px;  
}  
  
.form-group input:focus {  
    outline: none;  
    border-color: #667eea;  
}  
  
.btn-login {  
    width: 100%;  
    padding: 12px;  
    background: #667eea;  
    color: white;  
    border: none;  
    border-radius: 5px;  
    font-size: 16px;  
    cursor: pointer;  
    transition: background 0.3s;  
}  
  
.btn-login:hover {  
    background: #5568d3;  
}  
  
.error-message {  
    background: #fee;
```

```

        color: #c33;

        padding: 10px;

        border-radius: 5px;

        margin-bottom: 20px;

        border-left: 4px solid #c33;
    }

    .error-message ul {

        margin: 5px 0 0 20px;

    }

</style>

</head>

<body>

    <div class="login-container">

        <div class="login-header">

            <h2>Login Sistem</h2>

            <p>Silakan masuk ke akun Anda</p>

        </div>

        <?php if (!empty($errors)): ?>

            <div class="error-message">

                <strong>Error:</strong>

                <ul>

                    <?php foreach ($errors as $error): ?>

                        <li><?php echo htmlspecialchars($error); ?></li>

                    <?php endforeach; ?>

                </ul>

            </div>

        </div>

```

```
<?php endif; ?>
```

```
<form method="POST" action="">
```

```
<div class="form-group">
```

```
<label for="email">Email</label>
```

```
<input type="email" id="email" name="email"
```

```
value="<?php echo htmlspecialchars($email); ?>"
```

```
required placeholder="nama@email.com">
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="password">Password</label>
```

```
<input type="password" id="password" name="password"
```

```
required placeholder="Masukkan password">
```

```
</div>
```

```
<button type="submit" class="btn-login">Login</button>
```

```
</form>
```

```
</div>
```

```
</body>
```

```
</html>
```

Database Configuration - config/database.php

```
<?php
```

```
class Database {
```

```
    private $host = "localhost";
```

```
    private $db_name = "skpl_db";
```

```
    private $username = "root";
```

```

private $password = "";

public $conn;

public function getConnection() {

    $this->conn = null;

    try {

        $this->conn = new PDO(

            "mysql:host=" . $this->host . ";dbname=" . $this->db_name,

            $this->username,

            $this->password

        );

        $this->conn->setAttribute(PDO::ATTR_ERRMODE,
PDO::ERRMODE_EXCEPTION);

        $this->conn->exec("set names utf8");

    } catch(PDOException $exception) {

        echo "Connection error: " . $exception->getMessage();

    }

    return $this->conn;

}

}

?>

```

SQL Schema

-- Create users table

```

CREATE TABLE users (

    id_user INT AUTO_INCREMENT PRIMARY KEY,

```

```

nama VARCHAR(100) NOT NULL,
email VARCHAR(100) NOT NULL UNIQUE,
password VARCHAR(255) NOT NULL,
role ENUM('admin', 'user') DEFAULT 'user',
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
);

```

```
-- Insert sample user (password: password123)
```

```
INSERT INTO users (nama, email, password, role) VALUES
```

```

('John Doe', 'john@example.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'admin'),

```

```

('Jane Smith', 'jane@example.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'user');

```

Dashboard Page - dashboard.php

```
<?php
```

```
session_start();
```

```
// Check if user is logged in
```

```
if (!isset($_SESSION['logged_in']) || $_SESSION['logged_in'] !== true) {
```

```
    header("Location: login.php");
```

```
    exit();
```

```
}
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html lang="id">
```

```
<head>

    <meta charset="UTF-8">

    <title>Dashboard</title>

</head>

<body>

    <h1>Selamat Datang, <?php echo htmlspecialchars($_SESSION['user_nama']); ?>!</h1>

    <p>Email: <?php echo htmlspecialchars($_SESSION['user_email']); ?></p>

    <p>Role: <?php echo htmlspecialchars($_SESSION['user_role']); ?></p>


    <a href="logout.php">Logout</a>

</body>

</html>
```

Logout - logout.php

```
<?php
session_start();
session_destroy();
header("Location: login.php");
exit();
?>
```